Lecture 10: IaC & CI/CD

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Agenda

- AWS CloudFormation
- AWS CDK
- CI/CD: AWS CodePipeline
- Terraform
- AWS Certificates

CloudFormation

- A service that helps you model and set up AWS resources so you can spend less time managing recourses and more time focusing on applications.
- Infrastructure as Code (IaC): Maintain and track versions of your infrastructure in the same way you do with software.
- Consistency and Safety: Standardize infrastructure components used across organization, enabling configuration compliance and faster troubleshooting.
- Automation: Reduce manual processes and eliminate the need for custom scripts for resource deployment.

CloudFormation: Key Concepts and Components

- Templates: JSON or YAML format files that describe the AWS resources and properties
- Stacks: A collection of AWS resources that you can manage as a single unit.
- Change Sets: A preview of how proposed changes to a stack might impact your running resources.

CloudFormation - Yaml

```
AWSTemplateFormatVersion: '2010-09-09'
Resources:
 MyS3Bucket:
  Type: 'AWS::S3::Bucket'
  Properties:
   BucketName: my-example-bucket
   VersioningConfiguration:
    Status: Enabled
   PublicAccessBlockConfiguration:
    BlockPublicAcls: true
    BlockPublicPolicy: true
    IgnorePublicAcls: true
    RestrictPublicBuckets: true
   Tags:
    - Key: Environment
      Value: Production
```

AWS CDK

- The **AWS Cloud Development Kit (CDK)** is an open-source software development framework to define cloud infrastructure using familiar programming languages like TypeScript, JavaScript, Python, Java, and C#. It provides a higher-level object-oriented abstraction to define AWS resources imperatively rather than declaratively, as in AWS CloudFormation.

AWS CDK

- High-level abstractions
- Familiar programming languages
- Code reuse and modularity
- Integrations with IDEs and Tooling
- Automated Infrastructure management
- Seamless CloudFormation integration

AWS CDK

```
import * as cdk from '@aws-cdk/core';
import * as s3 from '@aws-cdk/aws-s3';

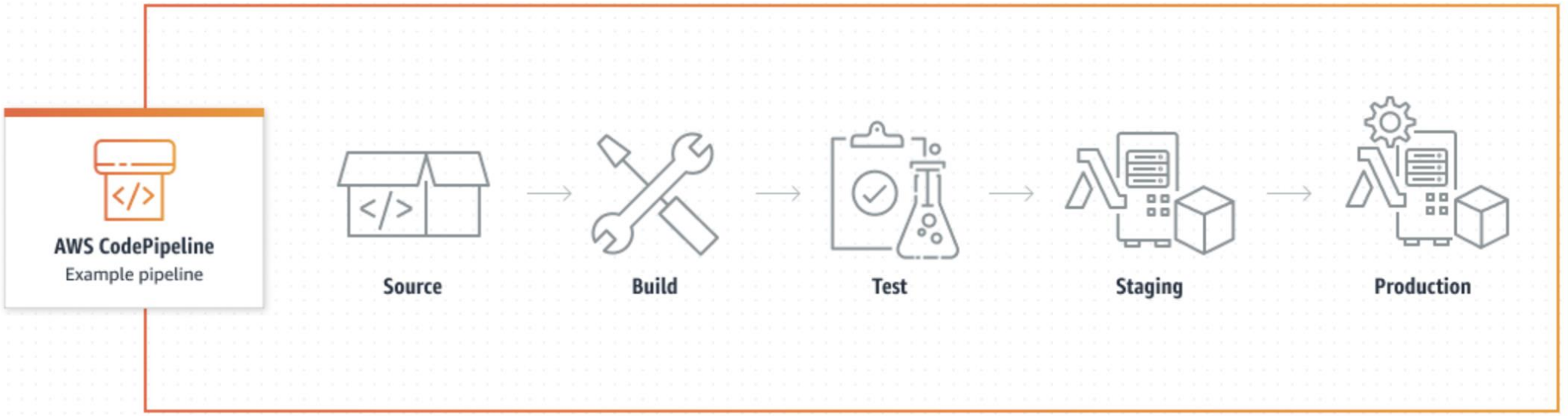

export class MyFirstStack extends cdk.Stack {
  constructor(scope: cdk.Construct, id: string, props?: cdk.StackProps) {
    super(scope, id, props);

    new s3.Bucket(this, 'MyFirstBucket', {
      versioned: true
    });
  }
}
```

YAML vs. CDK

YAML	CDK
Declarative	Imperative
Easy for non-programmers	Require knowledge of programming language
Limited for modularity	Easy for modularity with classes, functions
Use CloudFormation directly	Convert to CloudFormation template
Direct access to every resource	Higher abstraction

AWS CodePipeline



AWS CodePipeline

- CI/CD: Continuous Integration, continuous delivery
- CodePipeline: An AWS service for automating the steps required to release software changes continuously.
- Automation: Build, test, and deploy based on the code changes in the source repository
- Speed and Efficiency: Increase the speed of development cycles and improve efficiency by catching bugs early.
- Integration: Seamless integration with other AWS services (e.g., CodeBuild, CodeDeploy) and third-party services (e.g., GitHub, Jenkins).

AWS CodePipeline

- Key concepts:
  - Pipeline Model: Includes stages such as Source, Build, Test, and Deploy
  - Artifacts: Move between stages
  - Actions: Activities in a stage

Terraform

- Terraform is an open-source Infrastructure as Code (IaC) tool by HashiCorp.
- Automates the provisioning and management of cloud infrastructure.
- Cross-cloud-platform tool: AWS, Azure, GCP, Kubernetes, and over 100 other providers.
- Homepage: https://hashicorp.com/

Terraform

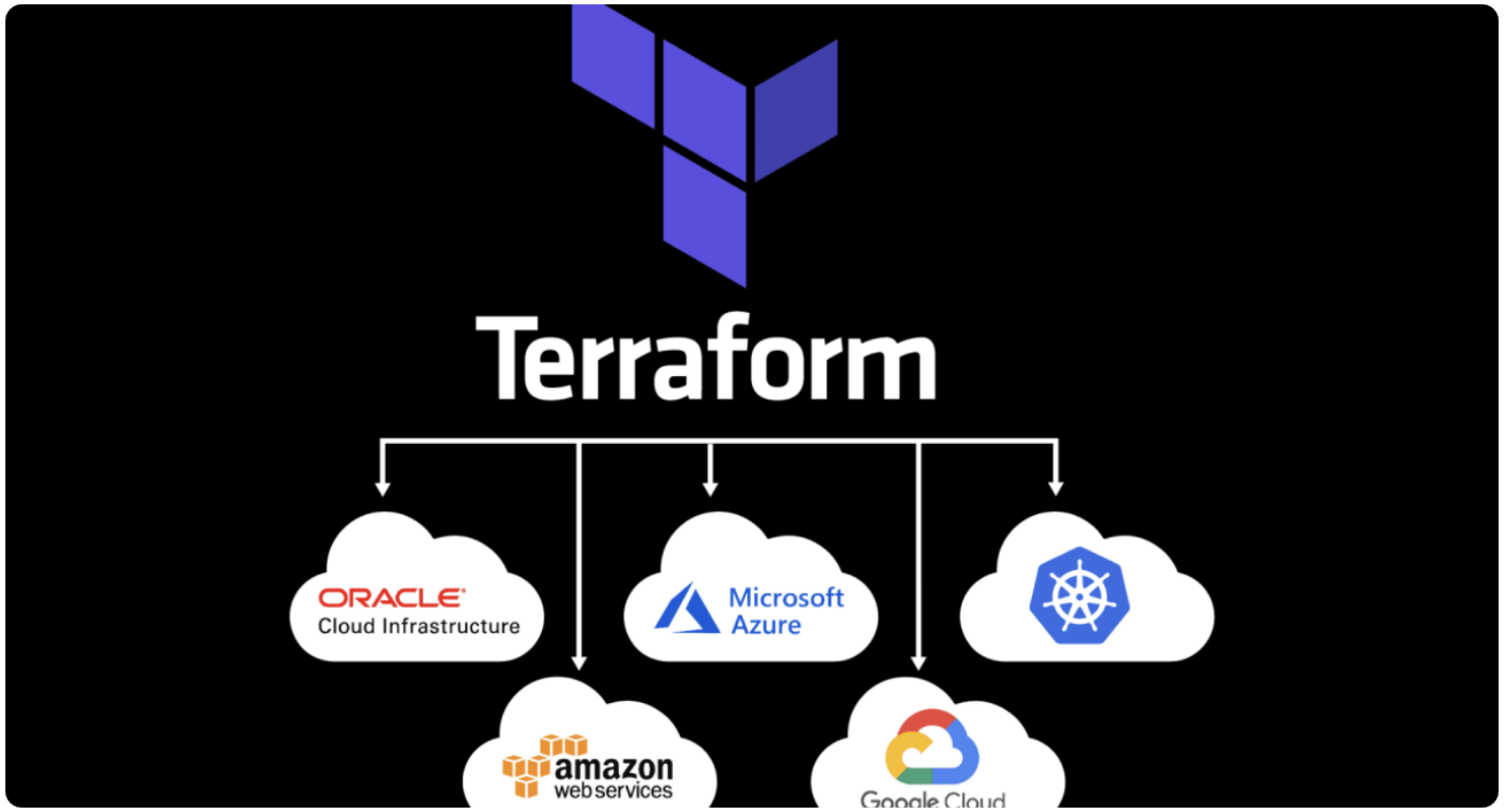


Image credit: https://k21academy.com/

Terraform

- **Multi-Cloud Support**: Write one configuration for multiple providers.
- **Infrastructure as Code (IaC)**: Manage infrastructure through reusable code.
- **State Management**: Keeps track of resources in a state file to prevent drift.
- **Modular Code**: Use modules to organize and reuse code across projects.
- **Version Control Integration**: Works well with Git, enabling collaboration and tracking.

Terraform

- HCL (HashiCorp Configuration Language) is a programming language that is designed to be easy to understand and write.
  - **Declarative Language**: Define **end states** (what you want) rather than **how** to achieve them.
  - **Modular Design**: Supports reusable **modules** for organizing infrastructure code.
  - **Variables and Outputs**: Use **variables** for flexibility and **outputs** for easy access to resource values.
  - **State Management**: Works with a **state file** to track changes to infrastructure.

Terraform

```
provider "aws" {
  region = "us-east-1"
}

resource "aws_instance" "web" {
  ami           = "ami-0c55b159cbfafe1f0"
  instance_type = "t2.micro"
}
```

Terraform

```
provider "google" {
  project = "my-project-id"
  region  = "us-central1"
}
resource "google_compute_instance" "vm" {
  name = "my-vm"
  machine_type = "e2-micro"
  zone = "us-central1-a"
}
```

Terraform

```
provider "azurerm" {
  features {}
}

resource "azurerm_linux_virtual_machine" "vm" {
  name     = "my-vm"
  location = "East US"
  size     = "Standard_B1s"
}
```

AWS Certificates for Developers

- AWS Certified Cloud Practitioner (Optional)
- AWS Certified Developer – Associate
- AWS Certified Solutions Architect – Associate
- AWS Certified Solutions Architect – Professional

AWS Certificates for DevOps

- AWS Certified Cloud Practitioner (Optional)
- AWS Certified SysOps Administrator – Associate
- AWS Certified Developer – Associate
- AWS Certified DevOps Engineer – Professional

AWS Certificates - Specialty

- AWS Certified Machine Learning – Specialty
- AWS Certified Data Analytics – Specialty
- AWS Certified Security – Specialty

AWS Certificates – Preparation Reference

- https://www.whizlabs.com/
- https://www.udemy.com/
- AWS Certification Training

References

- https://docs.aws.amazon.com/
- ChatGPT: https://chatgpt.com/
- Google AI: https://gemini.google.com/app
- https://hashicorp.com/